Artur Mrozowski

Senior Data Engineer / Data Architect

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Summary

Senior data architect and data engineering with 15+ years of experience designing source-of-truth data platforms, semantic data models, and operational analytics solutions across insurance, finance, and pharma.

Strong background in analytics strategy, data product platforms, metadata-driven automation, event-driven analytics, and cross-functional collaboration between business and engineering teams.

I specialize in turning complex business processes into governed analytical platforms that improve metric consistency, operational visibility, and decision support.

Core Competencies

- Analytics strategy and operational intelligence
- Data product platforms and enablement
- Semantic and canonical data modeling
- Unified business metrics and governed definitions
- Source-of-truth data platforms
- Event-driven analytics and operational modeling
- Metadata-driven automation
- Domain-driven design and Event Modeling
- Data governance, lineage, and metadata management
- Snowflake, SQLMesh, dbt, SQLGlot
- Kafka, GraphQL, Neo4j
- AWS and Azure

Experience

Senior Data Engineer, Tryg — Oct 2023 – present

- **Revenue Intelligence and Operational Analytics:**
- Developed a revenue intelligence platform enabling standardized business metrics including NDR, GRR, Logo Retention, and funnel analytics across domains.
- Built an event-driven analytical model enabling drill-down from executive KPIs to individual business events and operational decision points.
- Established reusable semantic definitions and time-series business events to improve metric consistency, explainability, and cross-functional trust in analytics.
- Enabled analysts and business stakeholders to trace revenue movements and customer lifecycle changes through sequenced business activities rather than isolated snapshots.
- Created a foundation for future context-aware analytics and richer operational intelligence capabilities.
- **Data Product Platform and Strategy:**
- Helped shape a data product strategy focused on reducing fragmentation, improving interoperability, and enabling scalable analytics delivery.
- Built a shared data product platform improving consistency, reuse, and delivery speed across analytical domains.
- Replaced fragmented self-service patterns with reusable governed building blocks and shared semantic concepts aligned with ACORD and domain-driven design principles.
- Replaced dbt with SQLMesh to improve scalability, maintainability, and developer workflow for analytical data products.
- Introduced isolated virtual development environments reducing dependency conflicts and improving collaboration across teams.
- **Metadata-Driven Automation:**
- Designed a metadata-driven model generation approach turning declarative YAML definitions into governed SQLMesh models.
- Automated generation of analytical building blocks including hook, bridge, and event models from reusable metadata, reducing manual SQL development and improving structural consistency across domains.
- Modeled temporal validity, dependency-aware bridge logic, and reusable business keys as metadata instead of hand-coded model patterns.
- Used dependency graphs and topological ordering to generate analytical structures in the correct sequence, improving repeatability and reducing delivery risk.

Senior Data Architect, Lundbeck — Sept 2021 – Sept 2023

- Defined a common data layer strategy reducing fragmentation and improving interoperability across analytics and operational domains.
- Replaced a monolithic data lake approach with a flexible platform inspired by Data Mesh principles.
- Enabled scalable self-service analytics while maintaining governance and semantic consistency.
- Collaborated across technical and business domains to align analytical capabilities with organizational needs.
- **Technologies:** Snowflake, dbt, AWS

Data Engineer / Architect, Tryg Aug 2016 – Aug 2021

- Built an underwriting data platform replacing Excel-based workflows with centralized governed data flows and shared business logic.
- Introduced Event Modeling to improve collaboration between business and engineering stakeholders.
- Designed enterprise semantic and canonical models based on ACORD to standardize concepts and improve interoperability across domains.
- Built event-driven write-back capabilities using GraphQL and Kafka, enabling applications to publish operational outcomes back into the analytical ecosystem.
- Automated logical and physical model generation using metadata-driven techniques, improving consistency and reducing manual effort.
- **Enterprise Canonical Data Platform:**
- Developed a semantic data platform enabling standardized interchange of business information across departments.
- Implemented graph-based semantic models in Neo4j aligned with the ACORD insurance ontology.
- Exposed semantic models through GraphQL APIs and integrated event streams using Kafka.
- Improved consistency and discoverability of enterprise data assets.
- **Metadata and Data Catalog:**
- Implemented metadata-driven lineage and business glossary capabilities using semantic technologies and Neo4j.
- Exposed metadata and lineage through Amundsen data catalog.
- Improved discoverability, governance, and understanding of analytical assets across teams.

Data Architect, KMD Nov 2015 – Aug 2016

- Designed target architecture and business proposal for a large-scale MS SQL Server data warehouse initiative.
- Participated in Data Governance Board activities defining enterprise data standards, policies, and governance principles.

Senior DW/BI Developer, Nordea Aug 2014 – Nov 2015

- Led evaluation and proof of concept for data warehouse automation tooling using WhereScape RED.
- Improved reliability and reduced time-to-market for BI deliveries through automation and streamlined delivery workflows.

Earlier Career

- Swedish Agency for Marine and Water Management — BI Consultant (2013 – 2014)
- DR — Business Intelligence Developer (2012 – 2013)
- Danske Bank — External Consultant (2010 – 2012)
- SimCorp — Software Developer (2008 – 2010)
- Copenhagen Energy — DBA (2004 – 2008)
- Quibus International AB — Software Developer (2002 – 2004)

Current Learning Professional Development

- Knowledge Graph Academy ongoing
- Claude Academy Claude Certified Architect track (ongoing)

Certifications

- Certified Scrum Product Owner (2022)
- Data Mesh — Domain Oriented Data (2022)
- Lead Enterprise Architecture Program on Azure — Microsoft (2017)
- Data Modelling Master Class — Steve Hoberman (2016)
- Data Vault Certified Data Modeler (DVCDM) (2014)
- CDMP Preparation (2024)

Education

Lund University

- Faculty of Informatics.
- Key areas: Database systems, data modeling, software architecture, UML, object-oriented design, Java programming, algorithms, and data structures.

Additional Education

- Apache Spark Workshop for Developers
- Distributed Computing with Spark
- Data Science Specialization — Johns Hopkins University
- Statistical Analysis with R
- Oracle BI EE
- C++ Programming
- SQL Server Administration
- Oracle Administration

Speaking and Open Source

- *Democratizing Data* — Tryg and Prisma (2021)
- *Building Semantic Data Hub with LPG, GraphQL and Kafka Streams* (2019)
- GraphQL & Graph Data Modeling in Neo4j
- Azure Functions & Machine Learning
- XML Parser for Danish Motor Register Data